Multi-Agent Cooperative Retrieval-Augmented Generation for Efficient Electric Power Knowledge Retrieval Quiz

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***Abstract*—In the context of electric power management, organizations often face the challenge of handling a vast amount of unstructured and complex documents. Traditional information retrieval systems struggle to provide precise, relevant answers to business-critical queries. This paper introduces a Multi-Agent Cooperative Retrieval-Augmented Generation approach, which leverages multiple LLMs to filter and assess the relevance of retrieved documents in a cooperative manner. The method enhances the performance of traditional RAG systems through collaboration between three specialized agents: the Forecaster, Evaluator, and Final-Forecaster. This framework reduces noise in the retrieved documents, ensures higher accuracy in answers, and improves the reliability of the system in electric power knowledge tasks.**

Keywords—multi-agent system, electric power knowledge, Retrieval-Augmented Generation, relevance score

I. Introduction

Power companies need to deal with a large number of complex documents and unstructured data in their daily operations, such as policies and regulations, financial statements, customer feedback, equipment operation and maintenance records, and so on. Facing diverse questions and business scenarios, traditional information retrieval methods are difficult to provide accurate and efficient answer support.

Large Language Models(LLMs) have demonstrated strong text generation, question and answer, and summarization capabilities in natural language processing tasks, and have played a large role in the field of power knowledge answering[1]. However, the knowledge of the LLMs itself is entirely derived from its training data. When these models are applied to deal with private or unseen text and document collections in model training, it is often difficult for the LLMs alone to efficiently retrieve relevant information from a large amount of text, so much so that it generates outdated or factually inaccurate information, which is known as the "illusion" of the LLMs[2].

To overcome this difficulty, Retrieval-Augmented Generation(RAG)[3] has been proposed, which combines information retrieval and text generation to enhance the generative capabilities of LLMs by retrieving relevant information from external knowledge sources, enabling them to answer questions about specific document collections.

Currently, there are two main types of RAG implementations: one is a training-based approach[4], which usually requires a large amount of computational resources and a long training time, despite its excellent performance; the other is a training-independent approach[5], which possesses higher efficiency and ease of use as a plug-and-play lightweight solution. However, the latter is highly dependent on the quality of retrieved documents, and the presence of irrelevant or noisy information not only reduces the accuracy of the answer, but also increases the computational overhead and affects the stability of the system[6]. Therefore, how to efficiently filter and sort the retrieval results has become a key issue that needs to be solved in the current RAG system.

To tackle this problem, this paper proposes a Multi-Agent Cooperative Retrieval-Augmented Generation(MAC-RAG)[4] based approach for electric power knowledge quizzing. Instead of relying on the traditional training or fine-tuning process, the

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method filters and scores the retrieved electricity documents through the collaboration between multiple large language model intelligences. Through the consensus mechanism between the intelligences, MAC-RAG is able to effectively identify the most relevant and highest quality documents for the generation process, thus significantly reducing the noise interference and improving the performance and reliability of the RAG system without sacrificing the recall rate.

II. Multi-Agent Cooperative Retrieval-Augmented Generation

A. Retrieval-augmented Generatione

RAG is an artificial intelligence technique that combines information retrieval techniques with language generation models. The technique works by retrieving relevant information from external knowledge bases and feeding it as Prompts to LLMs in order to enhance the ability of the models to handle knowledge-intensive tasks such as question and answer, text summarization, content generation, etc[8].

1) Retrieval: retrieval is the first step in the RAG process, retrieving information relevant to the problem from a pre-established knowledge base. The purpose of this step is to provide useful contextual information and knowledge support for the subsequent generation process.

2) Enhancement: Enhancement in RAG is the use of retrieved information as contextual input to a generative model (i.e., a large language model) to enhance the model's ability to understand and answer specific questions. The aim of this step is to incorporate external knowledge into the generation process to make the generated text content richer, more accurate and more responsive to user needs . Through the enhancement step, the LLM model is able to make full use of the information in the external knowledge base.

3) Generation: generation is the last step of the RAG process. The purpose of this step is to generate a response that meets the user's needs in conjunction with LLM. The generator uses the retrieved information as contextual input and combines it with the Large Language Model to generate textual content.

Fig. 1 is a representative instance of the RAG process applied to electric power knowledge question answering.

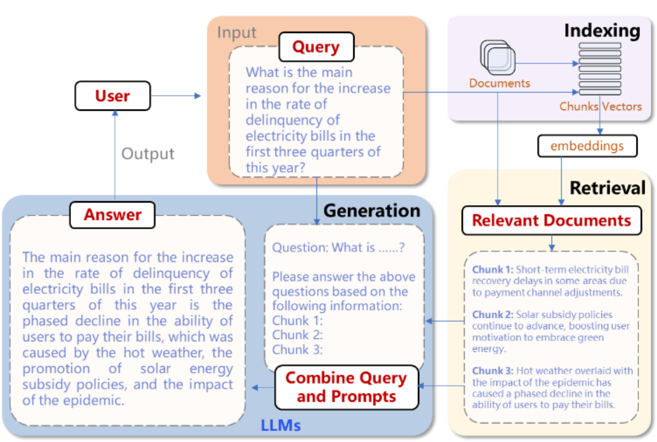

Fig. 1. A representative instance of the RAG process applied to electric power knowledge question answering

B. Multi-Agent Cooperative Retrieval-Augmented Generation

MAC-RAG is based on the framework of RAG and designs three LLMs agents with complementary functions to achieve recognition and filtering of noisy documents: Agent-1 (Forecaster), Agent-2 (Evaluator) and Agent-3 (Final-Forecaster).

Agent-1 (Forecaster): this agent starts working after document retrieval is complete, generating the corresponding answer for each candidate document of a query. Through this step, the system constructs the Document-Query-Answer (Doc-Q-A) triad, which provides input for the relevance assessment of the Evaluator Agent.

Agent-2 (Evaluator): based on the Doc-Q-A triad, this agent evaluates how well the document supports the query and the answer. The judgement process is reduced to a binary judgement ('yes'/'no'), a design that allows the relevance assessment to be quantified as a numerical score for subsequent document filtering and sorting.

Agent-3 (Final-Forecaster): after the Evaluator Agent completes the noisy document filtering and sorts the retained documents by relevance scores, this agent generates the final answer based on the processed document list.

The steps of the Multi-Agent Cooperative Retrieval-Augmented Generation approach are as follows:

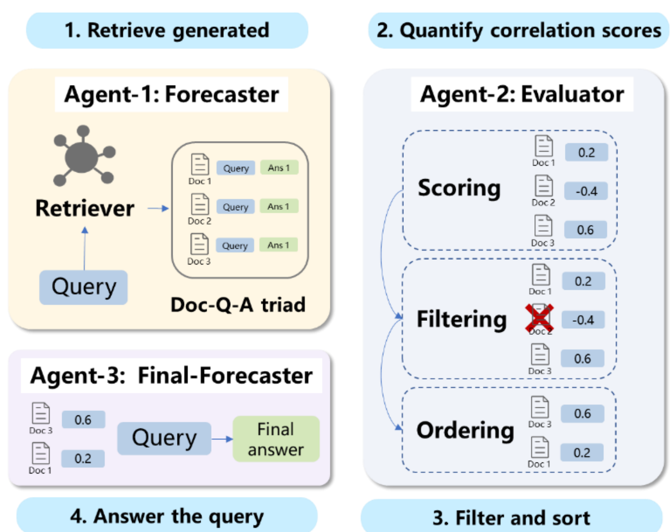

Fig. 2. The steps of the Multi-Agent Cooperative Retrieval-Augmented Generation approach

1) Retrieval to generate Doc-Q-A triad for relevance assessment.

a. In the document retrieval phase, a number of relevant candidate documents are retrieved for each user query. In order to improve the relevance coverage and ranking quality of the retrieved candidate documents, the Hybrid Retrieval Scheme is adopted, which scores the candidate documents returned from traditional sparse retrieval and semantic coding-based dense retrieval respectively, normalizes the scoring results, and introduces an adjustable parameter λ to control the weighting ratio of the sparse score S_{sparse} to the dense score S_{dense}. The final fusion score is calculated:

$$S_{\text{final}} = \lambda \cdot S_{\text{sparse}} + (1-\lambda) \cdot S_{\text{dense}} \tag{1}$$

Among them, $\lambda \in [0,1]$ can be set flexibly according to the system requirements, such as increasing the sparse weight in the scenarios requiring high keyword precision, and increasing the dense weight in the scenarios prioritizing semantic matching.

Finally, the set of candidate documents is sorted based on the fusion scores and the documents with the top scores are retained as the final candidate set for building the Doc-Q-A triad.

b. For a single query, the Forecaster Agent reasons out the answer to the corresponding query based on each relevant document, forming a Doc-Q-A triad. This triad is used as standardized input for the subsequent Evaluator Agent for relevance assessment and document quality judgement.

In particular, the triad generation process does not rely on any manual annotation or model fine-tuning, and is entirely based on the reasoning ability of the large language model itself, which has good training-free characteristics. That is, this step does not rely on task- or domain-specific supervised training data, but generates high-quality answers by directly understanding and inferring the semantic relationships between documents and queries through a general-purpose language model.

2) For a given Doc-Q-A triad, the Evaluator Agent evaluates and quantifies the correlation score between that document and the question and answer.

The specific steps for quantifying the relevance score of a triad are as follows:

First, for a given Doc-Q-A triad, the Evaluator Agent is instructed to judge whether the triad satisfies the following two conditions at the same time: (1) the document provides specific information for answering the question; (2) the answer to the large model is based on the retrieved document to answer the question directly. For each triad, if the above two conditions are satisfied, the answer is "yes"; if one of them is not satisfied, the answer is "no".

Secondly, in order to quantify the "yes" and "no" of the natural language output, the system uses forward propagation to pass the input text layer by layer to the last layer of the model during the inference process. The output layer, usually a fully connected layer. This layer maps the hidden representation into an output matrix with the dimension of the size of the vocabulary, representing the conditional probability distribution of each candidate token. At this point, the system extracts the "yes" and "no" tokens in the vocabulary and calculates their logits, and the difference of their logit probabilities is used as the relevance score of the document. The difference of the log probabilities represents the logarithm of the ratio of the probabilities of "yes" and "no", and by subtraction, the two factors are combined into a single score to simplify the judgement.

3) Adaptive thresholding based on relevance scores, filtering noisy documents and sorting documents based on their relevance scores.

After obtaining the relevance score of each document, a judgement mechanism based on adaptive threshold is proposed to effectively filter the noisy documents.[9] Specifically, the system sets the judgement threshold based on the average relevance score of the document collection corresponding to each query, so as to distinguish relevant documents from noisy documents. The scores of relevant documents are generally high and concentrated with small standard deviation, while the scores of noisy documents are more dispersed with large standard deviation, so the average score can be used as a more reliable judgement benchmark.

When the average score is high, it indicates that there are more relevant documents in the retrieval results, and the system will tend to eliminate low-scoring documents to exclude potential noise; while when the average score is low, the system appropriately retains a portion of intermediate-scoring documents in order to avoid omitting potentially relevant content, thus maintaining a high recall rate. In order to enhance the flexibility of this strategy, a weighted adjustment of the standard deviation σ is introduced, and the adaptive threshold is set to $\tau_s = \mu - n{\cdot}\sigma$, where is the average score of the document collection of μ , and n is an adjustable coefficient for dynamically adjusting the filtering criteria under different recall requirements.

4) After the Evaluator Agent has filtered out the noisy documents and sorted the list of remaining documents by relevance score, the Final-Forecaster Agent is prompted to answer the query using that list of documents.

The Final-Forecaster Agent generates answers based on the sorted document collection, combined with the user query, through the reasoning capability of the language model. The Agent discards redundant information interference and focuses on deep content fusion and semantic integration to output more accurate, concise and contextually appropriate answer texts. During the process of generating the final answer, the system supports a variety of output format controls, including summary statements, itemized lists, reference document positioning, etc., in order to enhance the user reading experience and trust. The final answer can be accompanied by supporting evidence fragments and corresponding document identifiers to enhance traceability.

III. Example Analysis

Introducing a Q&A system based on LLMs and multiple agent can help to improve the efficiency of electric utility operations and knowledge management. The following is an example of applying the solution to the operation and management of an electric power enterprise.

1) Retrieval to generate the Doc-Q-A triad for relevance assessment.

Firstly, the manager asks a query, for example, "What is the main reason for the increase in the rate of delinquency of electricity bills in the first three quarters of this year?". Second, the system calls the internal database and document management system to retrieve relevant documents, including customer service records, electricity bill collection reports, financial statements, user research reports, etc. Finally, for each retrieved document, the Forecaster Agent generates multiple structured Doc-Q-A triad based on a large language model reasoning about the possible answers to the query in the document.For example:

Triad 1

{
"document": {
"doc_id": "DOC_20250601".
"doc_name": "June 2025 Monthly Financial Report",
"source_type": "DOCX",
"content_snippet": "Short-term electricity bill recovery delays in some areas due to payment channel adjustments." },
"query": "What are the main reasons for the increase in the rate of delinquent customer electricity bills in the first three quarters of this year?" ,
"answer": "Due to the adjustment of payment channels in some areas, there is a time lag in the recovery of electricity bills."

Triad 2

{
"document": {
"doc_id": "DOC_20250602".
"doc_name": "2025 Green Energy Adoption Survey".
"source_type": "PDF",
"content_snippet": "Solar subsidy policies continue to advance, boosting user motivation to embrace green energy." },
"query": "What are the main reasons for the increase in the rate of delinquent customer electricity bills in the first three quarters of this year?" ,
"answer": "Solar subsidies boost user acceptance of green energy."

Triad 3

{
"document": {
"doc_id": "DOC_20250603".
"doc_name": "Q3 Customer Service Analytics Report 2025",.
"source_type": "PDF",
"content_snippet": "Hot weather overlaid with the impact of the epidemic has caused a phased decline in the ability of users to pay their bills." },
"query": "What are the main reasons for the increase in the rate of delinquent customer electricity bills in the first three quarters of this year?" ,
"answer": "Residents' ability to contribute is reduced due to high air conditioning usage due to hot weather and the impact of the epidemic on income."

2) The Evaluator Agent evaluates the relevance score between the document and the query and answer.

First, the Evaluator Agent evaluates the following two conditions for each triad: does the document provide specific information to answer the question? Does the answer come directly from the document ? For each field in the answer, the answer is "yes" if both conditions are met, and "no" if they are not.

Secondly, the logarithmic difference between the probabilities of "yes" and "no" judgements was calculated as the relevance score. The following table shows the correlation scores calculated from three triad.

TABLE I. THE RESULT OF CORRELATION SCORES FROM THREE TRIAD

Triad	Correlation Scores
Triad 1	0.2
Triad 2	-0.4(possible noise)
Triad 3	0.6

3) Adaptive threshold filtering and ranking based on relevance scores.

Firstly, the mean score and standard deviation of the document collection are calculated, with the mean score $\mu \approx 0.13$ and the standard deviation $\sigma \approx 0.41$; secondly, the adjustable coefficients are set at $n = 0.5$ and the adaptive threshold is $\tau_q = \mu - n\cdot\sigma \approx -0.08$, whereupon documents with scores lower than 0.08, such as triad 2, are excluded; lastly, the remaining documents are ranked in descending order according to relevance scores, generating a list of documents to be used for subsequent answer generation.

4) The Final-Forecaster Agent generates the answer.

First, the Final-Forecaster Agent reason using the sorted list of documents and user's query.

Inputs: document 3, document 1.

Output answer: "According to the third quarter customer analysis report and financial data, the main reason for the increase in the delinquency rate of users' electricity bills in the first three quarters of this year was the frequent use of air-conditioners due to the hot weather, coupled with the impact of the epidemic on the incomes of some residents and the decline in their willingness to pay bills. At the same time, the adjustment of payment channels in some areas also had an impact on the efficiency of payment."

Finally, the system outputs the answer and simultaneously returns metadata such as document provenance, time, relevance score, etc. for managers to refer to, improving decision-making transparency.

In addition, this study conducted a comprehensive evaluation of RAG and MAC-RAG models on the open-source long text question answering task dataset ASQA and a self built power dataset. We rely on the official metrics, including correctness with string exact match (str-em) and ROUGE score (rouge), and fluency measured by MAUVE (mau). Through experiments, it can be concluded that compared to RAG, MAC-

RAG has better accuracy and fluency in question answering tasks. The following table shows the experimental results.

TABLE II. EXPERIMENTAL RESULTS OF RAG AND MAC-RAG MODELS

Model	ASQA			power dataset		
	str-em	rouge	mau	str-em	rouge	mau
RAG	30.0	35.7	74.3	34.5	46.8	65.4
MAC-RAG	**39.2**	**42.0**	**76.6**	**48.9**	**52.7**	**70.1**

IV. CONCLUSION

The proposed Multi-Agent Cooperative Retrieval-Augmented Generation framework successfully addresses the limitations of traditional retrieval-augmented generation systems. By employing multiple agents that collaborate to filter out noisy documents and assess relevance, this method enhances the precision and reliability of answers to queries in the electric power knowledge. The system significantly improves decision support capabilities, enabling more efficient management and operational decision-making.

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